Diabetes related Hospital Readmissions: A Predictive Approach

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Abstract

This project looks at what drives 30-day hospital readmissions for diabetes patients and how well different machine learning models can predict who is most at risk. Diabetes continues to be a major cause of repeat hospital visits in the United States, which puts pressure on both patients and healthcare systems. Using the UCI Diabetes 130-US Hospitals dataset and the HRRP dataset, this project worked with both patient-level and hospital-level information to understand patterns behind readmissions. After cleaning the data and selecting useful features, several predictive models, including Logistic Regression, Decision Tree, Random Forest, and KNN were built and their performance was compared. PCA was also applied to see how the data grouped together and to reduce some of the noise. The results suggest that age, previous hospital visits, medication updates, discharge type, and overall visit complexity play a big role in whether a patient comes back within thirty days. At the hospital level, predicted and expected readmission rates were usually close, while excess readmission scores showed more variation. Overall, this project reinforces how helpful predictive analytics can be for identifying high-risk patients early and improving care transitions to prevent avoidable readmissions.

Project Introduction

Hospital admissions are a critical part of healthcare delivery across the world. In United States itself, millions of patients are admitted to hospitals every year for a variety of conditions. While the event of hospitalization is meant to improve patient outcomes, repeated visits indicate failures in the healthcare system. These failures could be at level of quality of care, patient education, or even post-discharge support. Readmissions are stressful for patients, costly for hospitals, and frustrating for families, making reducing readmissions increasingly important.

Among the conditions leading to hospital admissions, diabetes mellitus, commonly known as diabetes, stands out the most. It is a chronic condition that affects millions of Americans and is actually the leading cause of hospital readmissions. In 2018 alone, over eight million diabetic patients were discharged from U.S. hospitals, accounting for about 30% of all discharges (Kukde et al., 2024). While these readmissions reflect gaps in clinical care, they also contribute to the enormous financial burdens faced both by hospitals and patients. For example, Kukde et al. (2024) reported that hospitals spent roughly $12.8 billion managing endocrine-related complications in that year. On the other hand, for patients, the average cost of a 30-day unplanned hospital readmission is estimated to be $16,037 (Ghabowen et al., 2024).

The Hospital Readmissions Reduction Program (HRRP), introduced under the Affordable Care Act (ACA), penalized hospitals with higher-than-expected readmission rates for certain conditions, including heart failure, acute myocardial infarction, and pneumonia (Psotka et al., 2019; Dhaliwal & Dang, 2024). While the HRRP originally focused on just a few specific conditions, by 2018 it had grown to cover seven major conditions which impacted many hospitals. As a result, approximately 79% of hospitals that were participating in Medicare faced penalties (Psotka et al., 2019).

The penalties doled out by the HRRP were financially meaningful. Hospitals could lose up to 3% of their Medicare reimbursements for having excessive readmissions. Psotka et al., 2019 reports that the Medicare savings from the program increased from $290 million in 2013 to $564 million. This increase shows that even small reductions in hospital readmissions can lead to major savings for the hospitals and better results for patients.

This project will focus on predicting 30-day readmissions for diabetes patients. Diabetes readmissions often capture both medical and social factors, like age, race, income, insurance, and other health conditions (Kukde et al., 2024). As such looking at diabetes could help reveal patterns that we could apply to hospital readmissions on a broader scale. Additionally, a tangent of this project will also include examining HRRP data set to provide a system-level perspective on how hospital policies and performance metrics might influence patient outcomes.

Literature Review

Hospital readmissions, particularly for diabetes, have been extensively studied at both patient and hospital levels.

Policy and Financial Context

As mentioned in the introduction of the project, the Hospital Readmissions Reduction Program (HRRP) has made 30-day readmission rates a central quality measure. This program penalizes hospitals with higher-than-expected readmission rates, and by 2018, nearly 8 out of 10 hospitals were affected, with penalties reaching up to 3% of their Medicare payments (Psotka et al., 2019). Hospitals have worked hard to lower readmissions, but there are still concerns about fairness, equity, and patient outcomes. In fact, some hospitals try to avoid these penalties in ways that don't actually improve care, for example, moving patients to observation stays or changing how visits are coded (Psotka et al., 2019; Dhaliwal & Dang, 2024).

Patient and System Risk Factors

Patient characteristics tend to play a huge role in predicting readmissions. For diabetes patients, the risk factors include age, other health conditions like heart failure, length of hospital stay, discharge type, insurance type, household income, mental health, and race/ethnicity (Kukde et al., 2024; Cai & Islam, 2022). Social and economic factors like transportation challenges, housing instability, and low health literacy have also shown to increase readmission risk (Dhaliwal & Dang, 2024).

On the hospital's side, how care is organized also matters. Discharge planning, access to follow-up care, and coordination between healthcare teams all influence whether a patient comes back within 30 days. Hospital-level policies and metrics, such as those tracked by HRRP heavily interact with patient-level factors to shape outcomes (Psotka et al., 2019).

Intervention Strategies

There are some strategies that according to research do actually work to reduce readmissions. In fact, multidisciplinary approaches that involve doctors, nurses, diabetes educators, and pharmacists tend to be the most effective. Programs that combine structured discharge plans, early follow-up, patient education, and smooth care transitions have shown to consistently lower readmission rates (Cai & Islam, 2022; Dhaliwal & Dang, 2024).

For diabetes specifically, having certified diabetes educators, pharmacy-led education programs, and teaching patients basic “survival skills” during their hospital stay also helps reduce readmission risk (Cai & Islam, 2022). This goes to show that it’s not just the patient’s health that matters but also how the hospital educates and supports patients plays a big role too.

Machine Learning and Predictive Modelling for Readmissions

Predictive modeling has become an important tool for hospitals trying to anticipate which patients might be readmitted. A few recent studies have specifically focused on diabetes readmissions using the same UCI Diabetes 130-US Hospitals dataset that this project will use. These studies explored how different machine learning models could predict 30-day readmissions and identify the factors that most influence risk.

For example, Liu et al. (2024) compared ten traditional machine learning models, like Random Forest and Logistic Regression, alongside one deep learning model (an LSTM) to see which could best predict readmissions. They preprocessed the data carefully and selected the most important features using a technique called the Grey Wolf Optimizer. They also made sure the evaluation was fair by keeping all encounters from the same patient together in the same test or training fold. Their findings showed that Random Forest and XGBoost were consistently the strongest performers, achieving the best combination of accuracy, F1-score, and area under the curve (AUROC). The deep learning model performed well in some areas but did not outperform these tree-based methods on this dataset.

Similarly, another study by Emi-Johnson and Nkrumah (2024) compared four models: Logistic Regression, Random Forest, XGBoost, and Deep Neural Networks. They focused on both the accuracy and interpretability of the approached. Their goal was not just to predict readmissions but also to understand why the predictions were made. They used SHAP values, which is a method for explaining model predictions, to identify which features most influenced the risk of readmission. Across these studies, the key predictors consistently included discharge disposition, prior inpatient visits, age, comorbidities, and the number of medications a patient was taking.

The general consensus of these studies showed that tree-based models like Random Forest and XGBoost were powerful tools for predicting diabetes readmissions. They don’t just provide a yes/no prediction but also highlight the factors driving risk. This is important to note because it gives hospitals actual actionable information. For instance, if the model shows that a patient's discharge to a skilled nursing facility or multiple prior admissions increases readmission risk, clinicians can intervene with extra support, follow-up care, or education.

Data Understanding

This project uses two datasets:

This dataset contains over 100,000 patient encounters with information on demographics, diagnoses, lab results, medications, admission types, and discharge dispositions (Strack et al., 2014). It was collected from 130 U.S. hospitals between 1999 and 2008.
https://archive.ics.uci.edu/dataset/296/diabetes+130-us+hospitals+for+years+1999-2008

- While this data set is rich in-patient level data, it is slightly outdated, so some care practices might not reflect current standards. There are missing values, especially in lab results and medication records. There are also some categorical variables that will need to be modified for modeling.

Hospital-Level Data – HRRP Dataset

This dataset includes hospital-specific metrics on expected versus actual readmissions, excess readmission ratios, and penalties under the HRRP program. It provides a system-level perspective on how hospital performance and policies may affect patient readmissions.
https://www.johnsnowlabs.com/marketplace/hospital-excess-readmissions-reduction-program/

- The HRRP dataset also has some incomplete records for conditions, particularly if they did not meet the minimum case threshold. Additionally, the dataset is limited to hospital-level metrics, so it lacks patient-specific details, and coding practices tend to vary across hospitals. Despite these limitations, the HRRP data is still valuable for understanding system-level patterns.

Project Objectives and Expected Impact

Specific Aims

The project aims to predict which diabetes patients are most likely to be readmitted within 30 days and to identify the key factors driving these readmissions. A secondary aim is to also understand how hospital-level policies, such as HRRP penalties, might influence patient outcomes.

Expected Outcomes and Potential Impact

The goal of this project and the resulting model to identify high-risk patients before readmission occurs, giving hospitals actionable insights into providing targeted follow-up care, patient education, or additional support. By successfully pinpointing at-risk patients, this project can help reduce preventable readmissions, improve patient outcomes, and increase hospital efficiency.

At the same time, this project hope to shed light on how patient-level and system-level factors interact to influence readmission risk, offering insights that could inform broader healthcare strategies.

Data Cleaning and Pre Processing

Diabetes Dataset

The diabetes dataset needed some cleaning and preprocessing done before it could be used for analysis. Some columns were dropped because they had too many missing values, like A1Cresult, medical_specialty, payer_code, weight, and max_glu_serum, which ranged from about 39% to 97% missing. Others, like acetohexamide, tolbutamide, troglitazone, examide, citoglipton, and combination medications, were removed because they had very few unique values or were mostly dominated by a single value, adding little useful information. Unique identifiers like encounter_id and patient_nbr were also dropped since they did not seem to provide any meaningful information about readmission.

A new feature called total_prior_visits was created by summing number_outpatient, number_emergency, and number_inpatient, which gaves a better picture of a patient's overall prior healthcare use. After that, the original three columns were dropped to avoid any redundancy. Additionally, the target variable readmitted was also simplified from three categories to a binary outcome, with 'NO' as 0 and both '>30' and '<30' as 1, showing whether a patient was readmitted at all.

For imputation, missing values in race (about 2% missing) were filled with the most common category. Finally, all numeric features were normalized to reduce skewness and make sure everything was on the same scale for modeling to prevent any one feature from having a higher weight. Class weights were also applied to prevent the 'no readmissions' class from skewing the data.

Target Feature: Readmitted

Hospital Dataset

The hospital dataset needed some cleaning before it could be used for analysis. Columns that were mostly empty, like Longitude, Latitude, and Footnote_Description, and others that were redundant or not useful, like State_Abbreviation, Footnote_Code, ICD10_Description, HCPCS_Description, Start_Date, End_Date, Measure_Description, State_FIPS_Code, and Hospital_Name were also dropped.

For the remaining columns with missing values, numerical features like NPI, Number_of_Discharges, Excess_Readmission_Ratio, Predicted_Readmission_Rate, and Expected_Readmission_Rate were filled with the median to handle skew and outliers.

A new binary target, excess_readmission_flag, where hospitals with an Excess_Readmission_Ratio above 1.0 were labeled 1, and the rest 0, so the model just predicts whether a hospital has excess readmissions or not, was also created. Categorical features like State, Measure_Name, ICD10_Code, and HCPCS_Code were one-hot encoded with the first category dropped to avoid multicollinearity.

Finally, all numeric features, including the one-hot encoded columns, were scaled with StandardScaler to prevent any single feature from having a higher than actual weight when it came to modelling.

Target Feature: exces_readmission_flag

PCA

Diabetes Dataset

	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8	PC9	PC10
1	0.014037	-0.245541	1.286097	-0.984324	-0.329621	-0.843722	-1.053638	0.568082	-0.074504	0.768987
2	-1.044778	-0.368163	-1.010478	0.972538	-0.599826	-0.49132	1.324131	1.173752	1.093952	-0.87321
3	-0.91401	-0.261357	-0.278607	-0.737057	0.454999	-0.698625	0.120811	0.147142	0.242868	0.11716
4	-2.251351	-0.749579	0.078414	-1.350322	0.175143	-0.77068	0.292583	-0.45029	0.459944	0.135219
5	0.558644	-0.535136	-1.117186	1.091506	-1.501488	-0.706748	-1.032519	1.693178	0.902716	0.65472

Table 1.1

PCA Table

	encounter_id	patient_nbr	admission_type_id	discharge_disposition_id	admission_source_id	time_in_hospital	num_lab_procedures	num_procedures	num_medications	number_diagnoses	readmitted	total_prior_visits
PC1	0.171208	0.164013	-0.047404	0.106058	-0.041192	0.483526	0.362643	0.302282	0.535623	0.395461	0.109265	0.129203
PC2	0.569763	0.535995	-0.235788	-0.286394	-0.05545	-0.233593	-0.092475	-0.21162	-0.172485	0.249661	0.092293	0.197199
PC3	-0.228539	-0.120968	-0.042962	0.140931	0.495698	0.061967	0.154216	-0.425089	-0.124479	0.160152	0.473704	0.438008
PC4	0.052151	0.209024	0.721596	0.053758	0.132788	-0.121133	-0.435991	0.286975	0.131319	-0.013157	0.209451	0.252907
PC5	0.13871	0.183188	0.208771	0.221414	0.61214	0.028803	0.076951	-0.168489	-0.038677	0.224905	-0.528019	-0.337425
PC6	0.134642	-0.02483	-0.117195	0.821076	-0.346503	0.018302	-0.303287	-0.181516	-0.058685	0.135245	-0.087564	0.136458
PC7	0.051877	-0.256203	-0.102294	-0.1522	0.090974	-0.040182	-0.047269	0.097005	0.100033	-0.061021	-0.600345	0.708774
PC8	-0.080762	-0.324313	-0.342935	-0.100614	0.218588	-0.163331	-0.471506	0.339649	0.015678	0.556864	0.099188	-0.167965
PC9	0.079063	0.116452	-0.218656	0.354154	0.227644	-0.510698	0.375369	0.524365	-0.090658	-0.230453	0.113572	0.065048
PC10	-0.181138	-0.138329	0.376712	-0.044807	-0.355348	-0.39941	0.408091	-0.050679	-0.170797	0.553772	-0.105512	0.047534

Table 1.2

PCA Loadings

For the PCA, the focus was on understanding what each component actually represented once the original features were transformed. Each principal component is basically a weighted mix of the original variables; the strongest loadings were examined to see the patterns they captured.

PC1

PC1 mainly captures overall visit complexity. It increases when patients stay longer, take more medications, and have more lab procedures. So this component reflects how intensive or resource-heavy the hospital visit was.

PC2

PC2 is driven almost entirely by the ID columns. Because of that, it mostly represents technical variation in the dataset rather than any clinical pattern.

PC3

PC3 brings together admission source, pror visits, and readmission. This component reflects a patient's history in the system and their likelihood of coming back.

PC4

PC4 is dominated by admission type. It basically separates patients based on how they entered the hospital, like emergency, urgent, or elective.

PC5

PC5 connects admission source with not being readmitted. A high score here usually means the patient came from a certain source and did not return, so this component captures outcome differences tied to where the patient came from.

PC6

PC6 is almost entirely defined by discharge disposition. It reflects where the patient was sent after their stay, like home, another facility, or something else.

PC7

PC7 highlights patients with a lot of prior visits who were not readmitted this time. It separates stable chronic patients from those who return more frequently.

PC8

PC8 represents patients with many diagnoses but fewer lab procedures. It captures a shift between diagnostic complexity and lab volume.

PC9

PC9 reflects patients who had more procedures but a shorter hospital stay. It captures a more procedure-focused visit rather than a long, resource-heavy stay.

PC10

PC10 mixes diagnoses, lab procedures, and a specific admission source. It represents a smaller, more specific pattern tied to how patients arrive and how complex their case is.

Hospital Dataset

	PC1	**PC2**	**PC3**	**PC4**	**PC5**
0	-0.389	0.729825	-0.62745	0.342073	-1.439823
1	-1.299173	0.359029	-0.522547	0.373591	-1.574728
2	0.571444	-0.624266	-0.884721	0.115237	-1.173611
3	1.864017	0.384077	-0.660161	0.294081	-1.312783
4	-3.255954	0.913286	-2.00084	-0.353362	-0.332633

Table 2.1 PCA Hospital Data Set

	Provider_ID	NPI	Number_of_Discharges	Excess_Readmission_Ratio	Predicted_Readmission_Rate	Expected_Readmission_Rate	Number_of_Readmissions
PC1	-0.032243	-0.009216	0.20515	0.110236	0.617274	0.605935	0.443229
PC2	-0.000522	-0.062676	0.709719	-0.096827	-0.310172	-0.30439	0.542342
PC3	0.680433	-0.234606	0.01703	-0.673425	0.041778	0.150971	-0.060348
PC4	-0.047944	0.929012	0.058272	-0.357776	0.000473	0.056902	-0.000609
PC5	0.730524	0.279051	-0.004671	0.608347	-0.013452	-0.109651	0.078441

Table 2.2 Hospital Dataset PCA Loadings

The PCA for Hospital Dataset showed the following results:

PC1

It is basically the readmission rate dimension. When a hospital scores high here, it means both its predicted and expected readmission rates are high.

PC2

This reflects hospital size and volume. High scores point to hospitals that discharge a lot of patients and also have a high number of readmissions simply because of their scale.

PC3

This shows a contrast between Provider ID and excess readmission performance. A high score means a higher Provider ID paired with a lower excess readmission ratio. A low score tells the opposite story.

PC4

This is almost entirely driven by the NPI. It behaves more like an administrative identifier than a real performance factor, so there is not much interpretive value here.

PC5

This links higher Provider IDs with higher excess readmissions. Both move in the same direction, so a high score here points toward hospitals that not only have higher Provider IDs but also higher excess readmission ratios.

Methodology - Models

Logistic Regression

Logistic regression was used to predict the probability of a readmissions in the diabetes. This type of model works by calculating a weighted sum of all input features and applies a logistic function to transform this sum into a probability between 0 and 1. The model then uses a threshold to classify the outcome as 0 or 1. During training, it finds the feature weights that minimize the difference between the predicted probabilities and the actual outcomes using a method called maximum likelihood estimation.

Classification Decision Tree

A decision tree works by splitting the dataset based on feature values that best separate the classes. It works by essentially following a path of “if-then” rules, starting at the root node and then splitting the data recursively into branches. The tree chooses splits by measuring how well a feature separates the classes using some form of criteria. This then continues until stopping conditions are met, ex: a maximum depth or minimum number of samples in a leaf. Each leaf node then represents a predicted class. This model was used only in the diabetes data set

Random Forrest

Random Forest is an ensemble of many decision trees. I selected it because it reduces the risk of overfitting that a single tree can have. Each tree is trained on a random sample of the data with replacement (bootstrap sampling), and at each split, it only considers a random subset of features. Once all these trees are built, the model predicts a hospital’s class by taking a majority vote across all trees. A lot like averaging the votes of all the doctors in a room for a diagnosis instead of just relying on one. This averaging effect attempts to improve accuracy and make the model more robust to noise and outliers. This model was used in both diabetes and hospital data set.

K-nearest Neighbor

KNN classifies a data point based on its closest neighbors in the feature space. Distance measures like Euclidean distance were used to determine which points are the “nearest.” To predict the class (readmitted vs not readmitted) of a new patient, KNN finds the closest k points and assigns the most common class among them. This model was only applied to Diabetes Dataset, however due to the limitations of RAM capacity and KNN being lazy learning model, the data was sampled to reduce the computational load. The model is also sensitive to the scale of features, which is why the data was renormalized after sampling.

EDA

Diabetes Data Set

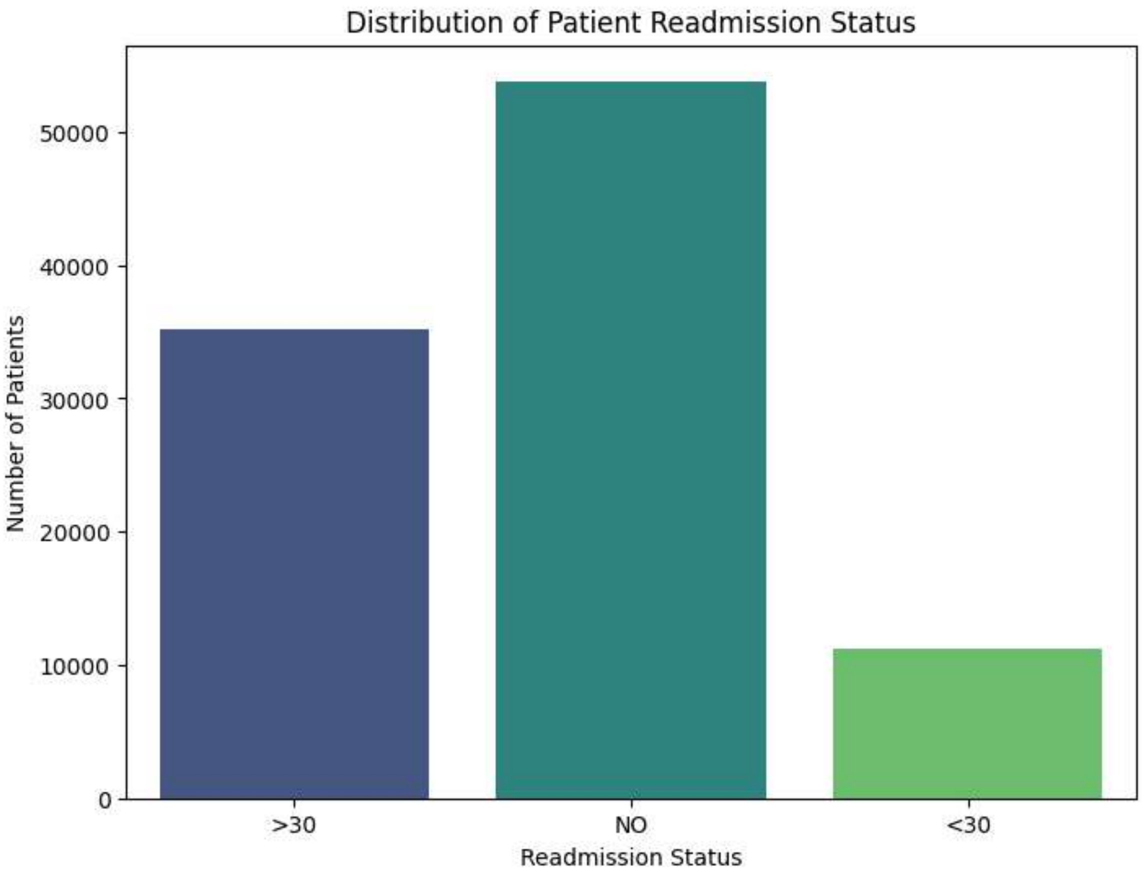
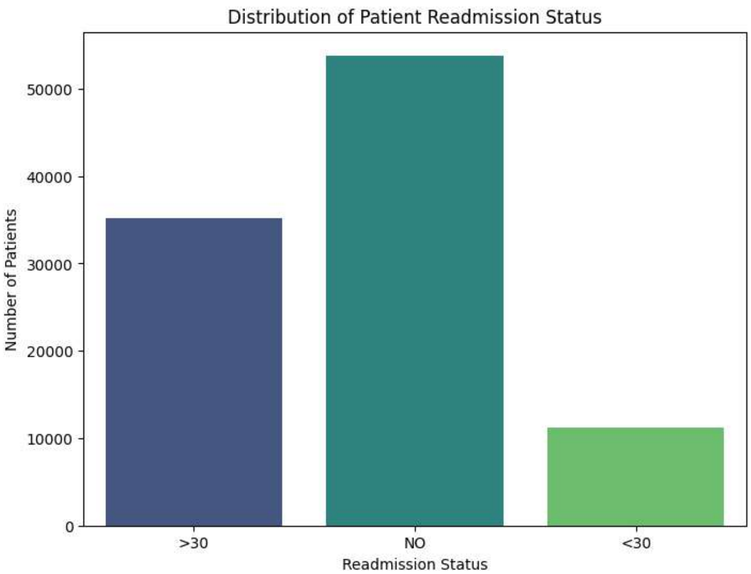


Figure 1.1

This plot shows that the sample has more ‘No’ readmission cases followed by >30 and then <30.



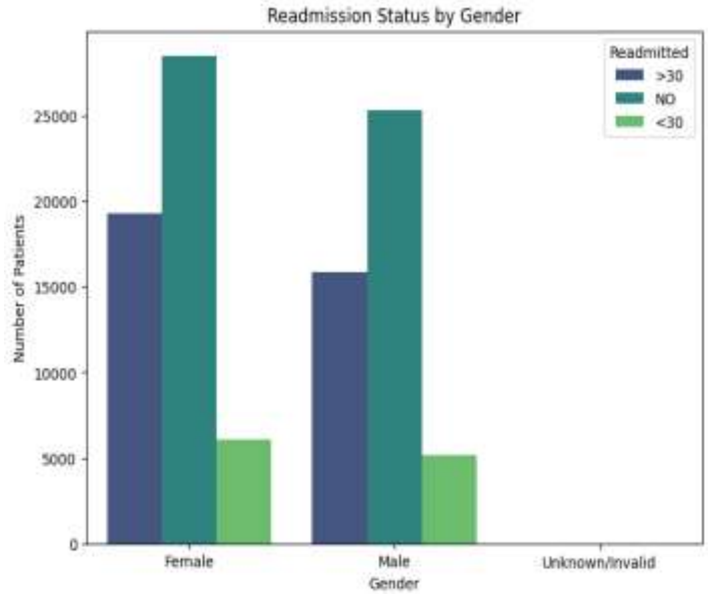
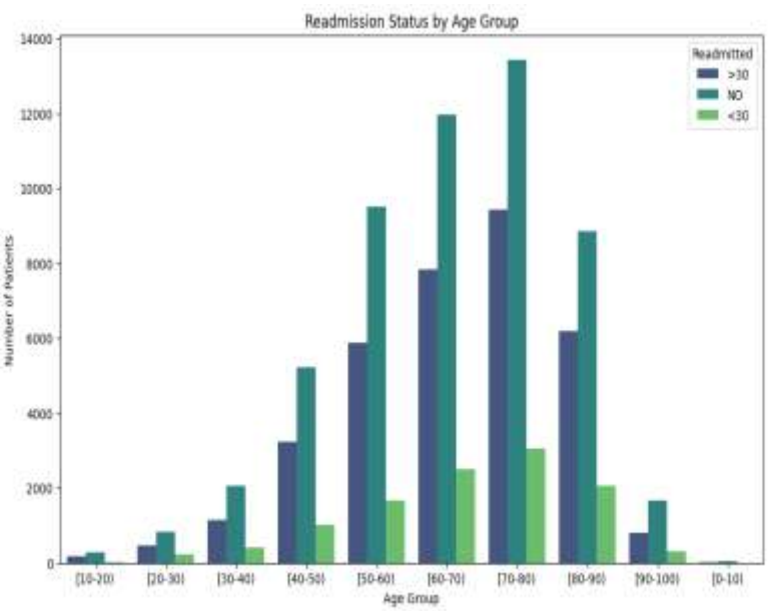


Figure 1.2

This bar chart shows that most patients were not readmitted, with smaller groups readmitted after 30 days and the smallest group returning within 30 days.

Figure 1.3

This chart compares readmission patterns across genders, showing trends for males and females, with most patients not readmitted, followed by those readmitted after 30 days, and the fewest returning within 30 days. Women show a higher representation in the sample.



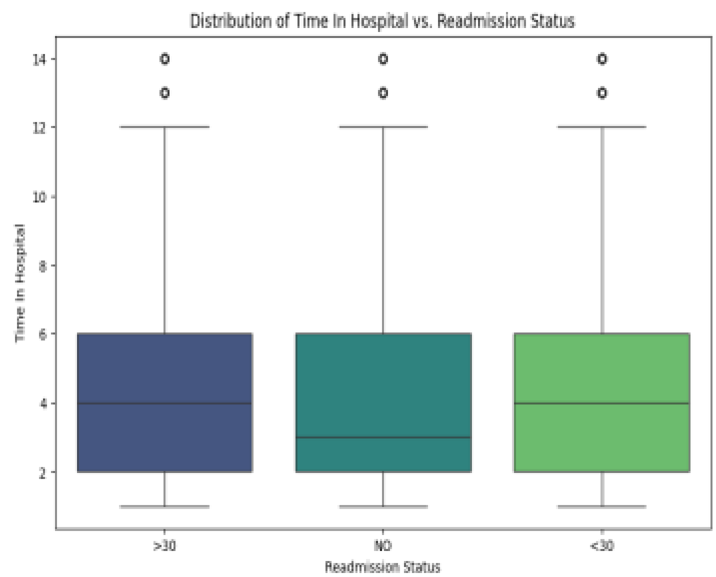
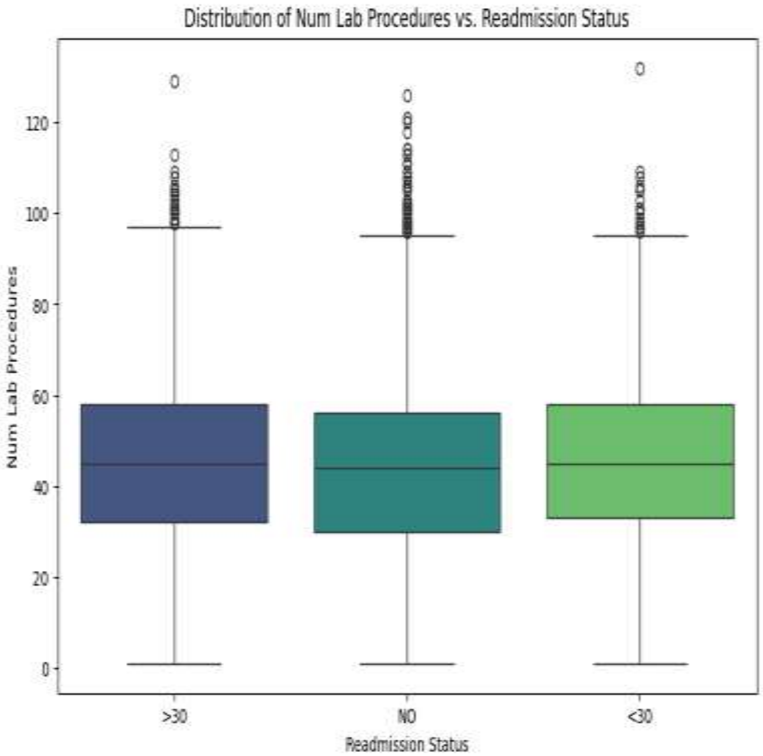


Figure 1.4

in cases through the 60–80 range suggests that age plays a significant role in shaping readmission outcomes.

Figure 1.5

This plot shows that time in hospital is still a useful variable, because even small differences in medians and spread can signal meaningful variation in how patients progress toward readmission. The slightly longer stays in the >30 group suggest that patients who require more inpatient care may still face elevated long-term



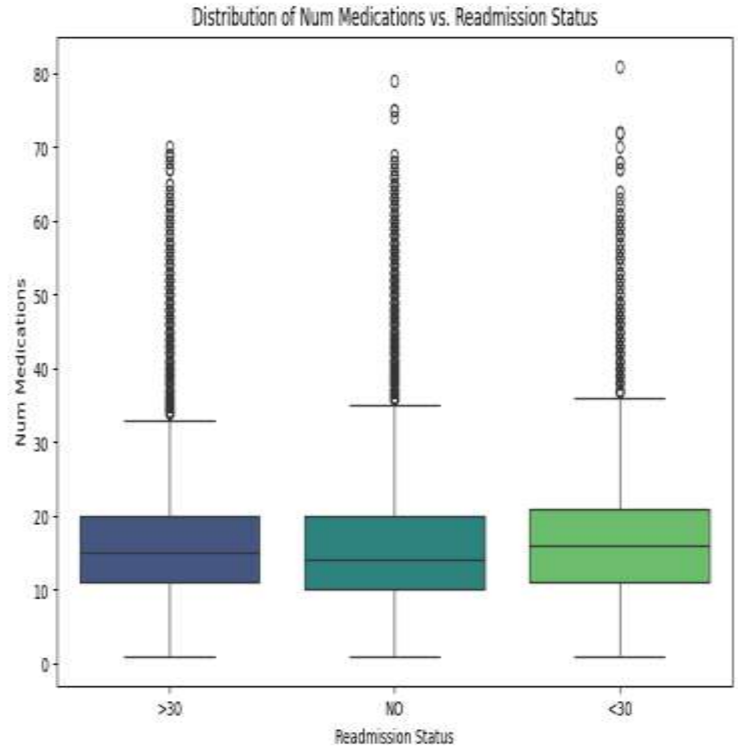


Figure 1.6

This plot shows that the number of lab procedures still has value for understanding readmission, because even with similar medians, patients in the <30 group show slightly higher overall testing intensity. This subtle increase suggests that patients requiring more lab work may have more complex or unstable clinical profiles, which can elevate their readmission risk.

Figure 1.7

This plot shows that the number of medications is a potentially useful indicator of readmission risk, since the <30 group trends slightly higher in medication count. The subtle upward shift suggests that patients managing more complex medication regimens may be more vulnerable to early readmission.

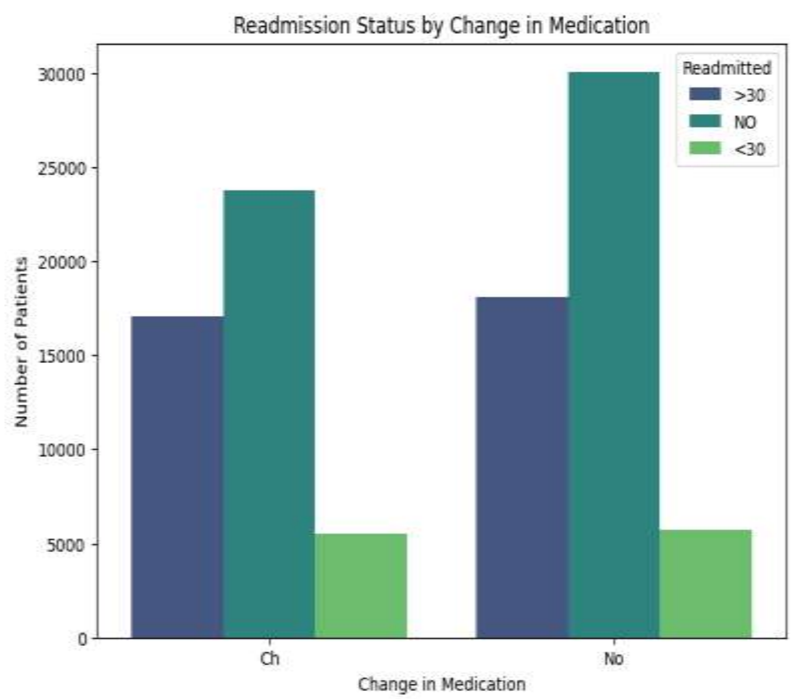


Figure 1.8

This plot shows that patients who had a change in their diabetes medication were much more likely to be readmitted, while those with no change were mostly not readmitted. The pattern suggests that medication adjustments signal higher clinical complexity, making these patients more vulnerable to both short- and long-term readmissions.

Summary

During the EDA, the first thing that stood out was the target variable. Most patients were not readmitted, and the smallest group came back within thirty days. This imbalance is important to note because models could naturally favor the majority class if it’s not addressed (this was addressed during Preprocessing and PCA).

Gender was fairly balanced, with women slightly outnumbering men, and both followed similar readmission patterns. While women in general tend to have higher rates of diabetes (Shi et al., 2025), this sample cannot conclusively prove that they also have higher rates of readmissions simply because they represent a higher number within the sample. Age was a clear predictor with older patients, especially those between sixty and eighty, making up most hospitalizations and readmissions, showing that age plays a meaningful role in readmission risk.

Looking at hospital stay and care, patients who were readmitted tended to stay a little longer, had slightly higher medication counts, and more lab procedures. These small differences suggest that patients with more complex care needs or more monitoring are more likely to come back. Medication changes were also important with patients whose diabetes meds were adjusted were much more likely to be readmitted, likely because these patients were already more unstable or complicated.

Prior healthcare use stood out as well. Patients with more outpatient visits, emergency visits, or past admissions were much more likely to be readmitted, especially within thirty days. Combining these into total prior visits made it even clearer that frequent prior care strongly predicts readmission.

Overall, the EDA showed that readmission is influenced by multiple factors: age, medication changes, the intensity of care during the stay, and prior healthcare use. These insights help explain what drives readmission and will guide how predictive models are built and interpreted.

Hospital Dataset

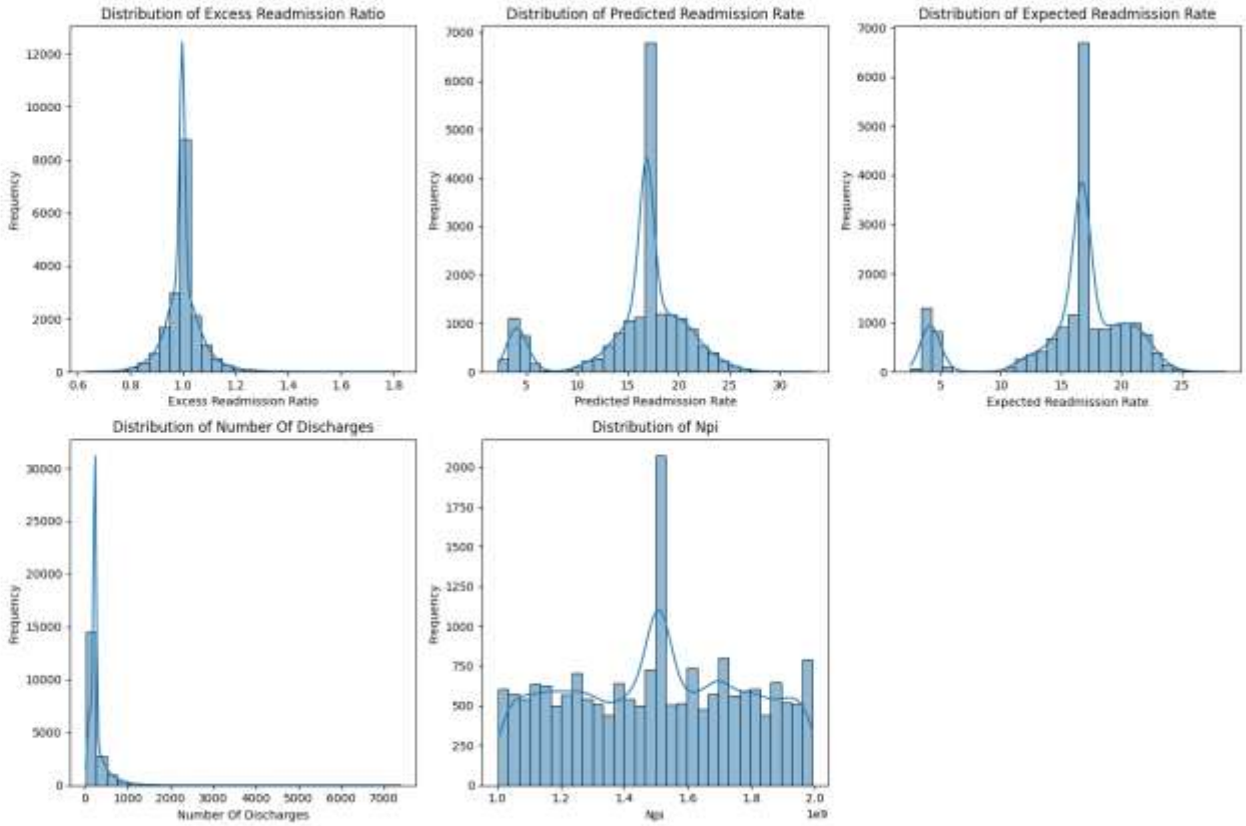


Figure 2.1

Excess Readmission Ratio:
Most hospitals are around a ratio of 1.0, which means their readmission rates are pretty much what was expected. So overall, most facilities are performing close to the standard.

Predicted Readmission Rate:
This one shows two peaks, around 5% and 18-20%, which tells us there are basically two groups of hospitals—some with low predicted risk and some with moderate risk.

Expected Readmission Rate:
It looks very similar to the predicted rate, confirming that the expected readmissions match the predicted pattern based on patient populations.

Number of Discharges:
Most hospitals have a low number of discharges, and only a few handle really high volumes.

NPI (National Provider Identifier):
This distribution is fairly even with a couple of peaks, which probably just reflects how the data was collected rather than anything clinical.

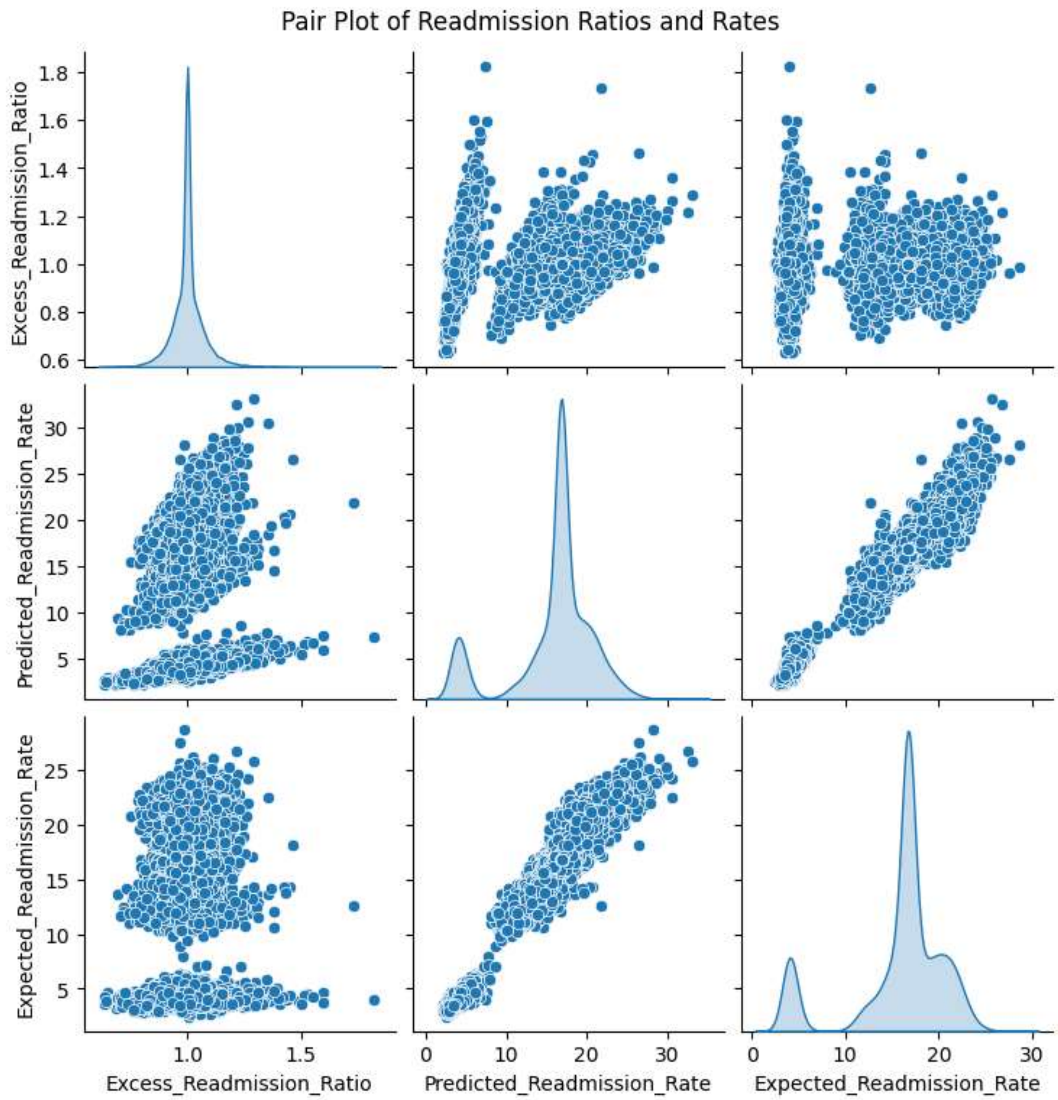


Figure 2.2

This plot shows how Excess Readmission Ratio, Predicted Readmission Rate, and Expected Readmission Rate relate to each other. The Predicted and Expected Rates are very strongly correlated, which means the model does a good job setting expectations based on historical data. The Excess Ratio only has a weak to moderate correlation with the other two, so a hospital's inherent patient risk only slightly predicts whether it will overperform or underperform. The diagonal plots confirm what we saw before: the rates are bimodal, while the Excess Ratio is centered around 1.0.

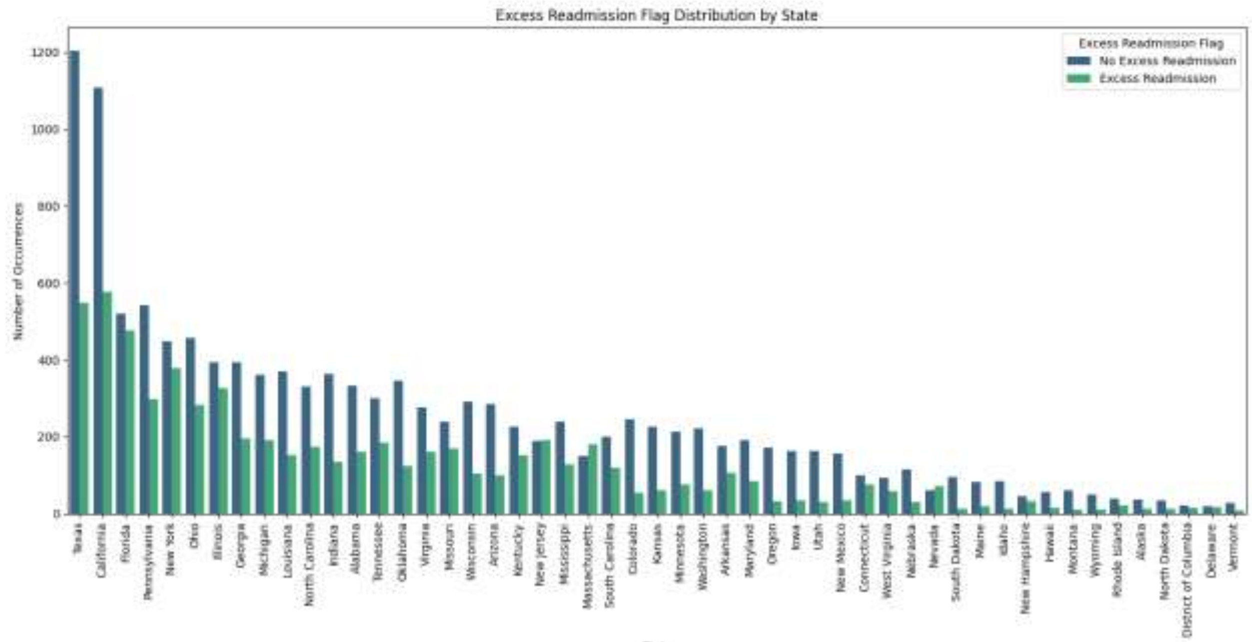


Figure 9.3

This chart shows how many hospitals in each state have an "Excess Readmission" flag versus a "No Excess Readmission" flag. Larger states like Texas, California, Florida, and Pennsylvania dominate the chart because they have the most hospitals. Overall, most states have more hospitals performing at or below the expected rate, though Texas and California also have the highest number of hospitals flagged for excess readmissions.

Summary

Looking at the EDA for hospital-level readmission data, it is apparent that most hospitals have an Excess Readmission Ratio around 1.0, which means their readmission rates are pretty much what was expected. The Predicted and Expected Readmission Rates show two peaks, suggesting there are basically two groups of hospitals: some with low predicted risk and some with moderate risk. Most hospitals have a low number of discharges, the NPI distribution seems to just reflect how the data was collected, and almost all hospitals fall under Footnote Code 5.

Looking at how these metrics relate to each other, the Predicted and Expected Rates are very strongly correlated, showing that the model does a good job setting expectations based on historical data. The Excess Ratio only has a weak correlation with the other two, so a hospital's patient risk only partly predicts whether it will over- or underperform. The distributions confirm what we saw before: the rates are bimodal, and the Excess Ratio is centered around 1.0.

At the state level, bigger states like Texas, California, Florida, and Pennsylvania dominate the chart because they have the most hospitals. In almost every state, more hospitals perform at or below the expected rate than above it, although Texas and California also have the highest number of hospitals flagged for excess readmissions. Overall, most hospitals are performing as expected, but there are clear patterns in predicted risk and some geographic differences that are worth keeping in mind.

Results

Diabetes Dataset

Model	Accuracy	Precision	Recall	F1-weighted	ROC AUC
Logistic Regression (Validation)	0.601982	0.607491	0.601982	0.602218	0.649267
Decision Tree (Validation)	0.582497	0.585776	0.582497	0.583048	0.62192
Random Forest (Validation)	0.630711	0.629246	0.630711	0.628089	0.676639
Random Forest (Test)	0.637295	0.636068	0.637295	0.634014	0.684251
KNN (Validation)	0.576011	0.575118	0.576011	0.575431	0.59783
KNN (Test)	0.582147	0.581526	0.582147	0.581772	0.607539

Table 3.1

Tabular Performance Metrics of all Models (Diabetes Dataset)

Actual Class	Predicted Class	Logistic Regression	Decision Tree	Random Forest	KNN
0 (NO)	**Predicted 0 (TN)**	4,595	4,609	5,706	4,250
	Predicted 1 (FP)	3,478	3,464	2,367	2,612
1 (Readmitted)	**Predicted 0 (FN)**	2,507	2,814	3,186	2,807
	Predicted 1 (TP)	4,457	4,150	3,778	3,112

Table 3.2

Confusion Matrix for all Models

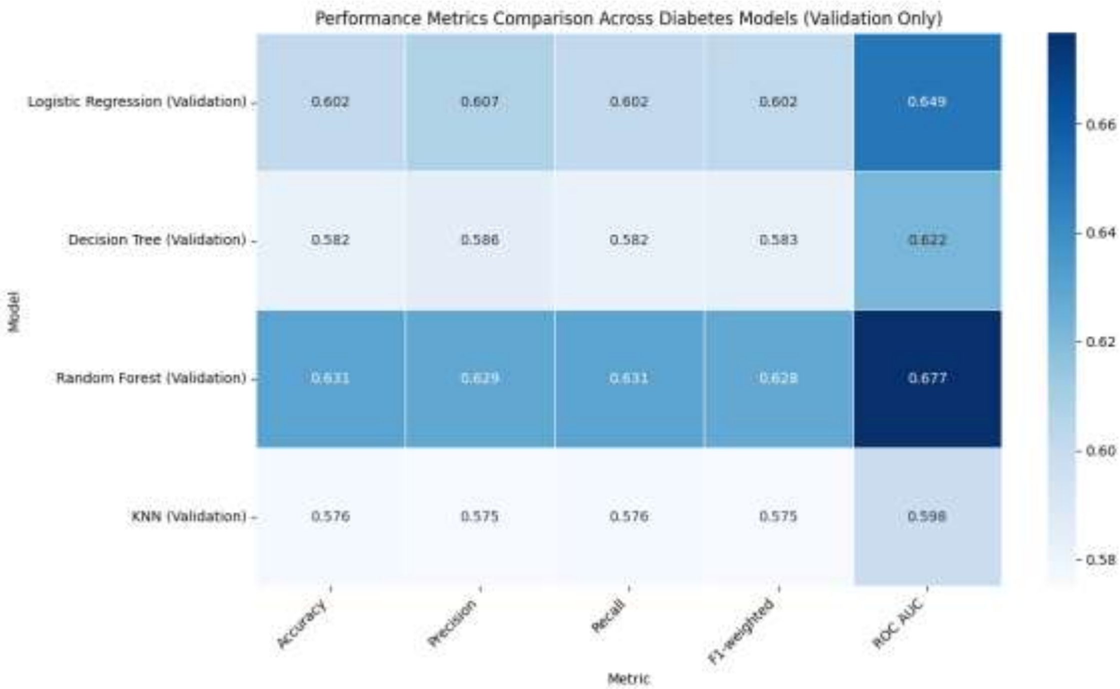


Figure 3.1

Heatmap for Diabetes Dataset Performance Metrics

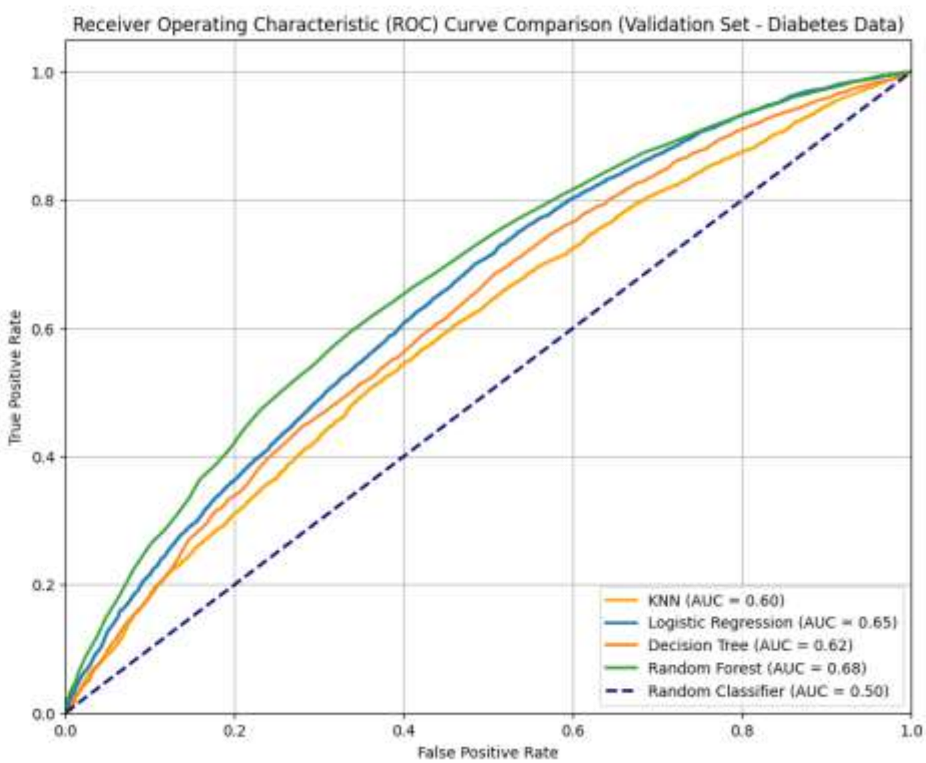


Figure 3.2

ROC AUC for Diabetes Dataset Models

Hospital Dataset

	Predicted Class 0	Predicted Class 1
Actual Class 0	1877 (True Negatives)	0 (False Positives)
Actual Class 1	0 (False Negatives)	976 (True Positives)

Table 3.3

Confusion Matrix for Random Forest

Metric	Value
Accuracy	**1**
Precision (weighted)	**1**
Recall (weighted)	**1**
F1-Score (weighted)	**1**
ROC AUC	**1**

Table 3.4

Performance Metrics for Random Forrest Hospital Dataset

Discussion

Diabetes Dataset

After running the dataset through multiple models and looking at the results, a clear pattern has emerged about how well different approaches handled the complexity of predicting readmissions in the diabetes dataset. While some models were able to pick up deeper, non-linear patterns that existed within patient history, hospitalization characteristics, and clinical variables, others unfortunately could not because the dataset simply demanded a richer structure than what the model could provide.

This section will compare and contrast the performance metrics, confusion matrices, and ROC AUC scores across all models that will paint a complete picture of how each model behaved and what its strengths and weaknesses were.

Random Forest

Random Forest was the strongest model overall. It achieved the highest accuracy and F1 scores (Validation: 0.63, Test: 0.64) and the top ROC AUC (Validation: 0.68, Test: 0.68). These scores indicate that it was able to strongly discrimination between readmitted and non-readmitted patients.

When we look at the confusion matrix, the model is also able to correctly identify 5,706 true negatives (the highest among all models) and 3,778 true positives. Its false positives (2,367) and false negatives (3,186) were also lower than most alternatives, showing that it balanced both classes effectively. Overall, Random Forest not only captured complex patterns in the data but also generalized well across both validation and test sets.

Logistic Regression

Logistic Regression as a model for this datast showed moderate performance with a validation accuracy of 0.60, F1 of 0.60, and ROC AUC of 0.65. At first glance, these numbers look promising, but the model's simplicity limits its ability to capture the non-linear and interactive effects in diabetes readmission.

Evaluating it's confusion matrix, we can see that it produced 4,595 true negatives and 4,457 true positives, but it also had 3,478 false positives and 2,507 false negatives. This high error rate, especially in misclassifying non-readmissions as readmissions, reinforces that while Logistic Regression identifies general trends, it misses many of the nuanced patterns in the data.

Decision Tree

The Decision Tree surprisingly had lower performance, with an accuracy and F1 around 0.58 and ROC AUC at 0.62. Its confusion matrix showed 4,609 true negatives and 4,150 true positives, but also 3,464 false positives and 2,814 false negatives. This shows that the single tree could not generalize well and it struggled to capture enough of the underlying structure. When we compare its performance to Random Forest, the results reinforce why ensemble methods are often superior for complex datasets.

K-Nearest Neighbors (KNN)

KNN performed the weakest overall. Its accuracy and F1 were around 0.58 on the test set, and it had the lowest ROC AUC at 0.60. The confusion matrix shows that it correctly identified 4,250 true negatives and only 3,112 true positives (the fewest among all models). Its False positives (2,612) and false negatives (2,807) indicate that it struggled with both classes.

Part of this underperformance is due to the algorithm itself being sensitive to high dimensionality and mixed feature types, which makes distance-based predictions less reliable. On top of that, KNN was only run on a sampled subset of the dataset to reduce computational demands, which likely limited its ability to fully capture the overall patterns in the data.

Comparing All Models

Overall, Random Forest clearly outperforms the other models. It balances precision and recall well while capturing complex, non-linear patterns in the data. Logistic Regression may seem competitive at first glance, but its simplicity prevents it from fully representing the complexity of diabetes readmission, which is influenced by multiple interacting factors such as comorbidities, medication regimens, and patient behaviors.

The single Decision Tree underperforms because it is too rigid and cannot generalize effectively across diverse patient profiles. KNN, on the other hand, struggles the most, partly because distance-based methods are sensitive to high-dimensional, mixed-type healthcare data but also because it was run on a sampled subset of the data.

It's important to recognize that, on a medical level, diabetes itself is a highly complex disease. These results too, show that accurate readmission prediction, especially for diabetes, needs models that are capable of handling multiple interacting variables and non-linear relationships. When we look at all the models within that context, Random Forest stands out as the most suitable choice, offering both strong predictive power and stability across validation and test sets.

Hospital Dataset

The Random Forest model did really well on the hospital dataset. In the most recent run, it got perfect score of 1.0 across accuracy, weighted precision, and recall. The confusion matrix showed 1,877 true negatives and 976 true positives, with no false positives or false negatives. This means the model was able to perfectly separate hospitals with excess readmissions from those without.

When the google notebook was executed a few days earlier for the poster presentation, the results were slightly lower but still very high. Accuracy was 0.9926, and the confusion matrix had a few misclassifications: 3,740 true negatives versus 13 false positives, and 1,924 true positives versus 29 false negatives. ROC AUC and other metrics were also near-perfect.

It is possible that the small difference between the two runs is probably due to randomness in model training or minor differences in preprocessing with the PCA-transformed features. Even with that, both runs show that Random Forest can capture strong patterns in the hospital data.

It is also worth noting that some features are closely related to the target feature in this dataset, which might make performance look a little better than it would in a real-world prediction scenario. Still, these results do show that Random Forest is a very good model foridentifying hospitals with high readmission risk.

Synthesis of Diabetes and Hospital Dataset

Understanding hospital readmissions is not straightforward, and looking at both patient-level diabetes data and hospital-level readmission data can give a much clearer picture- which has been my aim for this project.

The diabetes data shows individual patient risk factors that can increase the chance of readmission, like age, the number of prior outpatient, emergency, or inpatient visits, and changes in medication or diabetes treatment. For example, a patient who has had many prior visits and complex medication adjustments is naturally more likely to be readmitted because of their health complexity and treatment needs. When we combine these individual risks across patients, they directly contribute to a hospital's overall readmission performance, like the Excess_Readmission_Ratio. Hospitals that treat more high-risk patients may have higher readmission numbers, even if their care quality is good.

On the other hand, the hospital-level data gives me the bigger picture by summarizing patterns across the entire institution. Metrics like Predicted_Readmission_Rate and Expected_Readmission_Rate can help us understand whether a hospital is performing better or worse than expected, given the types of patients it serves. A hospital with a high Excess_Readmission_Ratio might indicate gaps in care coordination, discharge planning, or post-discharge follow-up that aren't fully meeting patient needs. Together, the patient-level and hospital-level data allow me to see both the individuals at risk and the broader systems or processes that influence outcomes.

Combining these datasets also highlights gaps in the readmission process. Some hospitals may not be fully addressing the needs of high-risk patients, while others may have inconsistencies in discharge planning, follow-up care, or post-discharge support. Certain groups of patients, like elderly diabetic patients with complex medication regimens, may require more focused interventions that aren't yet standardized. These insights can guide many of the potential practical strategies like hospitals improving discharge planning, care coordination, and staff training, while resources can be prioritized for high-risk patients and units with higher readmission rates. Patient-level interventions like personalized follow-up, home visits, telemonitoring, and culturally sensitive education are also important.

Overall, integrating patient-level and hospital-level analyses gives a much more complete and actionable view of readmissions. The models show who is at risk and highlight systemic gaps in hospital processes

Limitations

There are some important limitations to this analysis. Both the diabetes and hospital datasets don't include much economic or socio-economic information, which could affect readmission risk. The diabetes dataset covers 130 US hospitals from 1999 to 2008, so it may not fully represent today's patients or all regions.

Many variables, like lab results and medical specialties, had missing values or had to be dropped, which may have removed useful information.

For the hospital dataset, only Random Forest model after PCA was applied, so other modeling approaches weren't explored. The KNN model on diabetes dataset was run on a sample of the dataset for computational reasons, which may have limited its performance.

Recommendations

To improve predictions, more detailed clinical information, like lab results, medication dosages, and comorbidities, could be added for the diabetes dataset. Socio-economic and geographic data could also help identify external factors affecting readmissions. For hospitals, adding features like bed size, staffing levels, financial data, and EHR adoption could give a clearer picture of factors driving excess readmissions.

Additionally, using more advanced models, such as Gradient Boosting, stacking, or neural networks, could capture complex patterns better. Automated feature engineering could also reveal some hidden relationships. Finally, looking at patient journeys over time, investigating root causes of readmissions, and analyzing geographic hotspots could further help design more targeted interventions and policies. Another could be tracking these changes through the years in step with any Policy changes that were made.

Data Ethics

Working with healthcare data comes with a real ethical responsibility in addition to adherence to HIPPA laws. These data points aren't just numbers, they're tied to actual people, as such, protecting patient privacy has to be a priority, even in an academic setting; diabetes or the hospital dataset used in this project do not contain any identifiable information.

This project would also like to acknowledge just how easy it is for predictive models to pick up on bias without anybody noticing. If the data reflects existing inequalities in healthcare, the model can unintentionally reinforce them. Because of that, it's important to check for fairness, question unusual patterns, and make sure the model isn't favoring or disadvantaging any group.

In the end, the author of this paper agrees that being mindful of privacy, bias, and fairness is just as important as getting good accuracy, especially when the goal is to actually help patients and healthcare systems.

Conclusion

Overall, this project helped me get a clearer picture of what drives hospital readmissions, both at the patient level and the system level. By looking closely at diabetes patients and hospital level data, it became obvious that readmissions aren't caused by one single factor; they come from a mix of clinical complexity, prior healthcare use, social factors, and the way hospitals organize care. The models, especially Random Forest, showed real potential for identifying high-risk patients before they return, which could give hospitals a chance to step in with better follow-up and support.

Bringing both datasets together also highlighted how patient-level risk and hospital-level performance interact. Even with limitations in the data, the patterns were consistent: early intervention, strong care coordination, and better patient education can make a meaningful difference.

In the end, this project reinforced the idea that predictive modeling can be a powerful tool for improving patient care. If used thoughtfully, these models can help hospitals reduce preventable readmissions, ease financial burdens, and ultimately support better health outcomes for patients who need it most.

Citation

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